

Don't Panic

A Hitchhiker's Guide to
Agentic DevOps

Jeremy Meiss

**DON'T
PANIC**



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AI is an amplifier.

DevOps Research and Assessment (DORA), 2025

n = 5,000

CI throughput increase is deceptive

+59%

CI throughput, YoY

28M workflows analyzed - CircleCI 2026

Top 5%

where almost all of it sits

median team +4% · bottom quartile +0%



Our three takeaways

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1. Why some teams are pulling ahead while most aren't
2. What the actual bottleneck is now - it has moved, and it has a name
3. A set of steps you can bring to your VP on Monday that fit in a Jira ticket rather than a strategy deck

"Mostly Harmless"

Where teams actually sit

01

AI-Assisted

Copilot, Cursor, ChatGPT for the odd question. 72% use it daily (Sonar 2026). Four tools per team, no governance.

02

AI-Optimized

An actual decision got made - strategy, tool selection, security review. Roughly the top quarter.

03

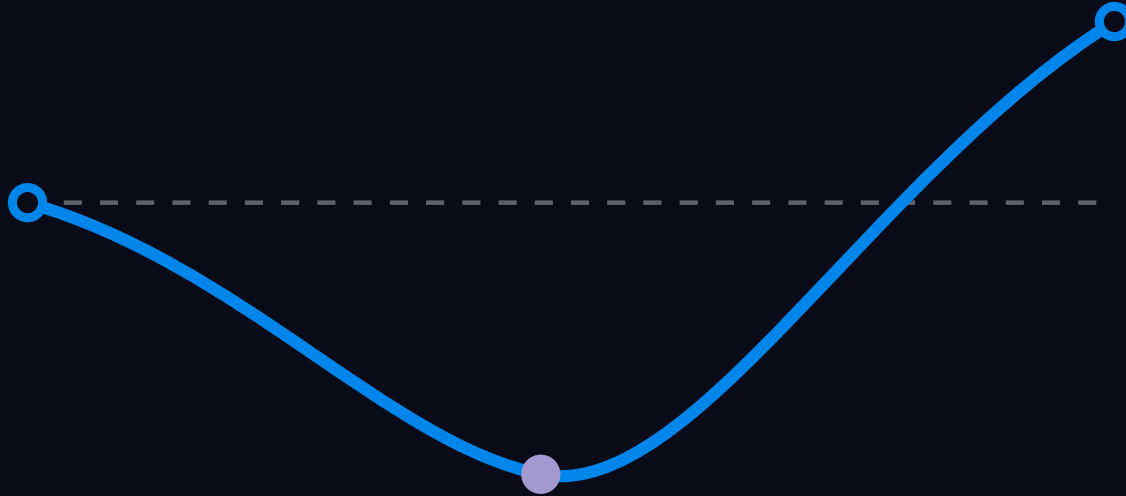
Agent-Augmented

Agents doing real end-to-end work. The human moves from in the loop to over the loop. The top 5%.

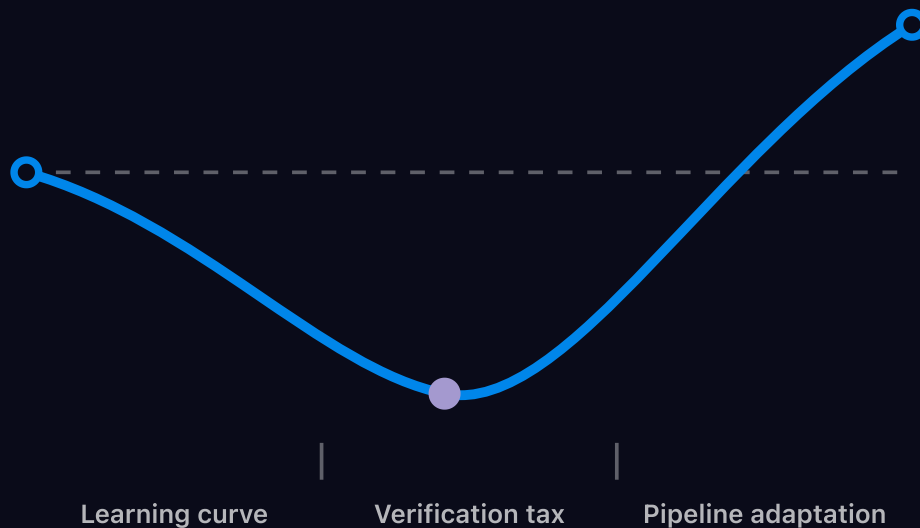




It isn't a straight line



The dip has a name



The tuition cost, not the failure.

Three reports, three conclusions

Google Cloud

OPTIMISTIC

78% of executives have seen ROI on at least one use case.

Stanford AI Index

NEUTRAL

Adoption is everywhere. Real structural change is rare.

MIT NANDA

PESSIMISTIC

The shadow AI economy - official tools underdeliver, people route around them.

Worth naming honestly

DORA's 2026 ROI report has taken real criticism this year - some call it more brochure than research.

Separately: Stride Research has **pre-registered** a study testing whether AI adoption still correlates with performance once you control for company size. Results expected later this year - not published yet.

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Be as skeptical of the AI-optimism industrial complex as you are of the AI-hype industrial complex.

It isn't just DORA

Atlassian, independently this year: the **AI efficiency paradox** – individual output speeds up, then backs up at review and approval.

Different company, different survey. No connection to DORA.

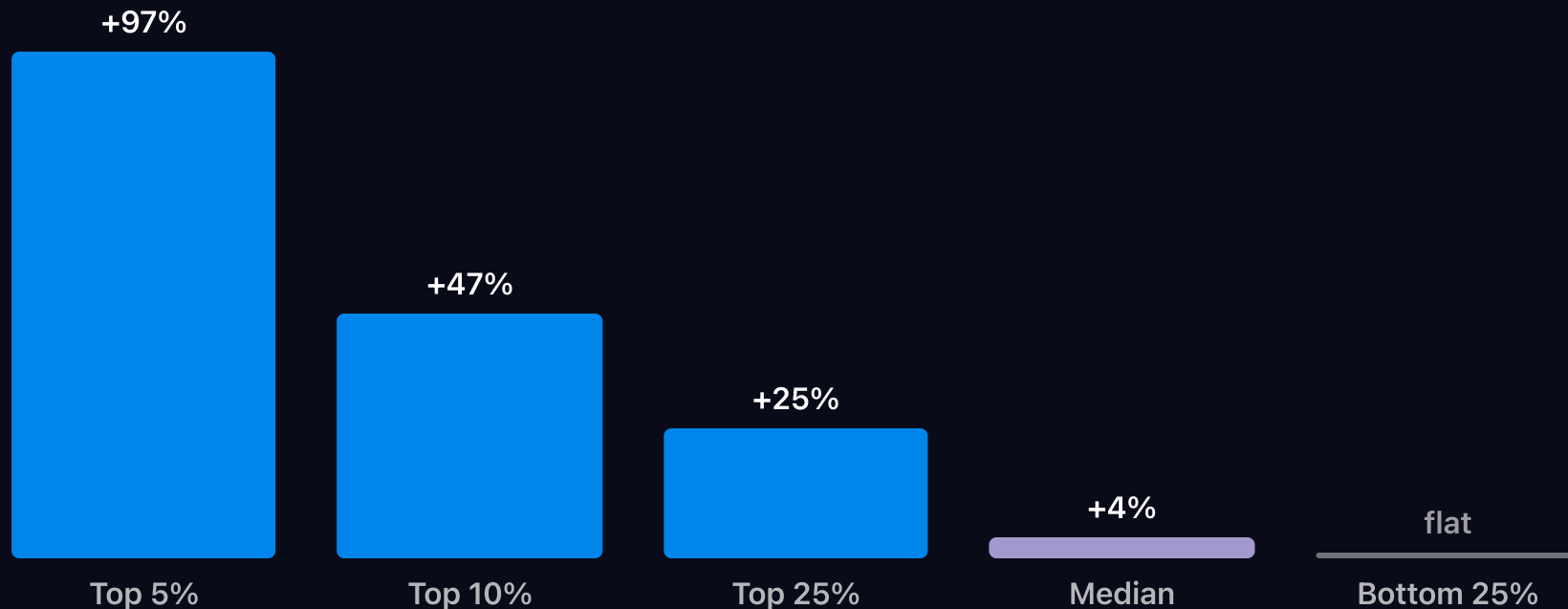
Not a survey

28,000,000

CI workflows analyzed

CircleCI 2026 - not a survey, actual pipeline data

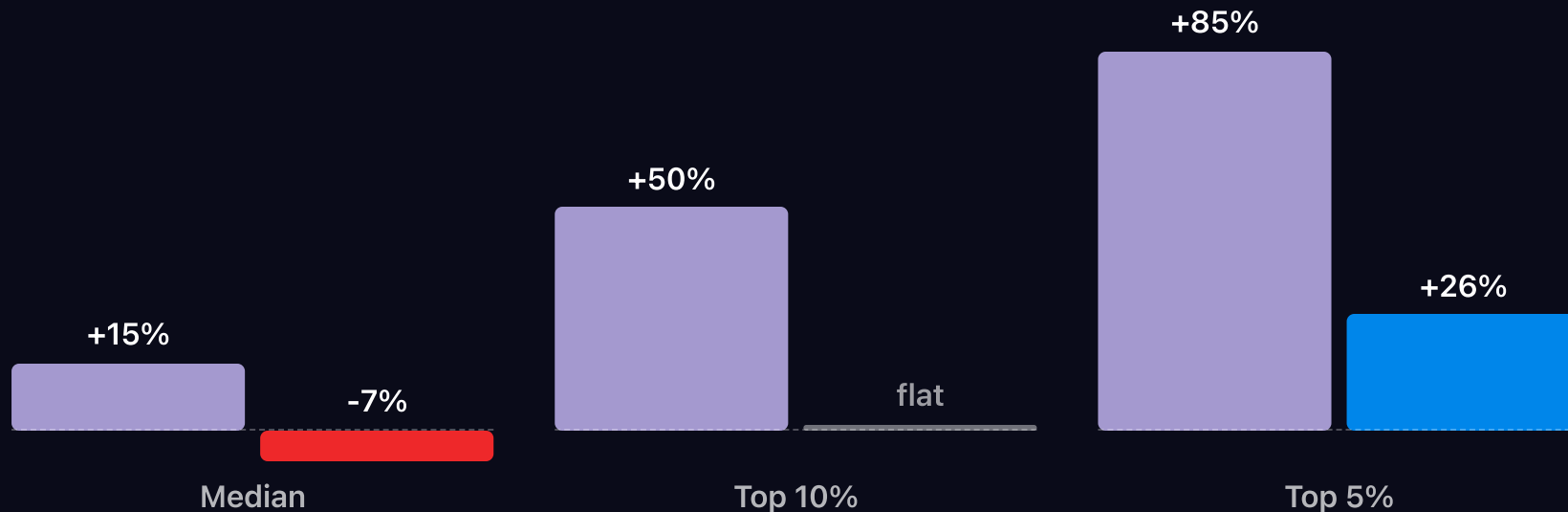
Throughput growth by percentile



CircleCI 2026

The bottleneck moved

■ Feature branch ■ Main branch



CircleCI 2026

Slower at fixing what breaks

72 min

Median time to recover

+13% YoY · feature branches: 80 min (+25%) - CircleCI 2026



Main-branch success rate

70.8%

September 2025

Lowest in 5+ years · benchmark is 90%

76.7%

March 2026

Improving - still short of the 2023–24 mid-80s and the 90% benchmark

Merge Efficiency Ratio - validation cycles per merged change

3.9

Median team

2.6

Top 5%

1.3

Elite cohort

20 organizations · CircleCI, July 2026

50-developer team, ~3,000 changes a month

\$900K

a year at risk

Unoptimized workflows - agents idling until their context cache expires

\$700K+

recoverable

By moving routine validation earlier - CircleCI, July 2026



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Throughput is a vanity metric. The real metric is integrations that ship.

Tokenmaxxing

Claudeconomics leaderboard - "Token Legend" tier

Meta: 85,000 employees. 60 trillion tokens. One month.

Uber: entire 2026 AI budget gone. Month 4 of 12.

Amazon, weeks later: leaderboard deleted.

"Don't use AI just to use AI."

Reported across multiple industry outlets, 2026 - figures not independently verified.



The gap is where incidents come from

96%

don't fully trust AI code

48%

always verify before
committing

Sonar 2026, n=1,149 developers



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Code that looks correct but isn't reliable.

61% of developers

Sonar, 2026

Predicted, felt, measured

Predicted: **+24% faster**

Felt: **+20% faster**

Measured: 19% slower

METR, randomized controlled trial, July 2025 - 16 developers, 246 real tasks.



2026: they tried to re-run it

METR tried to re-run the study in early 2026 to see whether newer tools changed the result - and the study itself broke.

Too many developers refused the "no AI" condition, even for pay. There was no longer a clean control group left to recruit.

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AI proposes. Humans dispose.

Every AI output is a junior dev's first PR.



Seven archetypes, six groups



DORA 2025, n=5,000

Add it up

38%

in clear difficulty

clusters C1 + C2 + C3

40%

high-performing

clusters C6 + C7 - DORA 2025, n=5,000

The shadow AI economy

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AI tools per team

35%

access via personal accounts

ChatGPT 52% · Perplexity 63% - Sonar 2026

The experience gap - Sonar 2026

Juniors

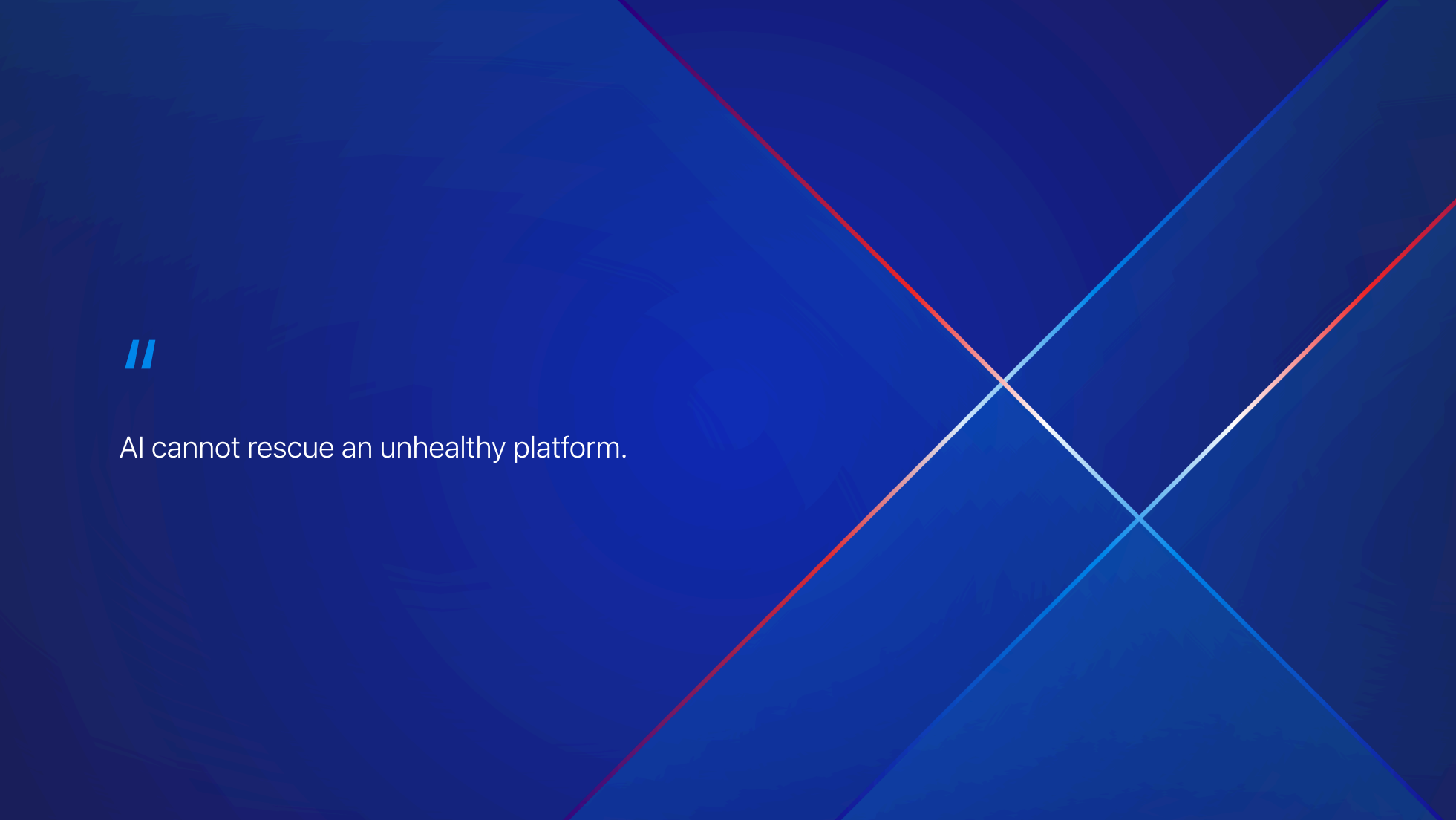
+40% productivity gain

66% report "looks correct but isn't reliable"

Seniors

+32% productivity gain

48% report "looks correct but isn't reliable"



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AI cannot rescue an unhealthy platform.

If it can happen there...

Dec 2025

Elevated permissions

Bypasses two-person approval, deletes and rebuilds the environment. 13-hour outage.

Mar 2, 2026

AI-assisted changes

120,000 lost orders. 1.6 million website errors.

Mar 5, 2026

Worse

~6.3M lost orders, a 99% drop in North American order volume, ~6 hours down. Amazon disputes AI as the primary cause.

Amazon's fix

Mandatory review, reinstated

Senior-engineer sign-off on AI changes. A human, back in the loop.



Amplifier works both directions.

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(Vibe) Build, then verify.

Generate fast. Validate harder.

Self-healing test automation (vendor-reported figures)

Before

Coverage 30–50%

Authoring takes days to weeks

Payback in 12–24 months

After

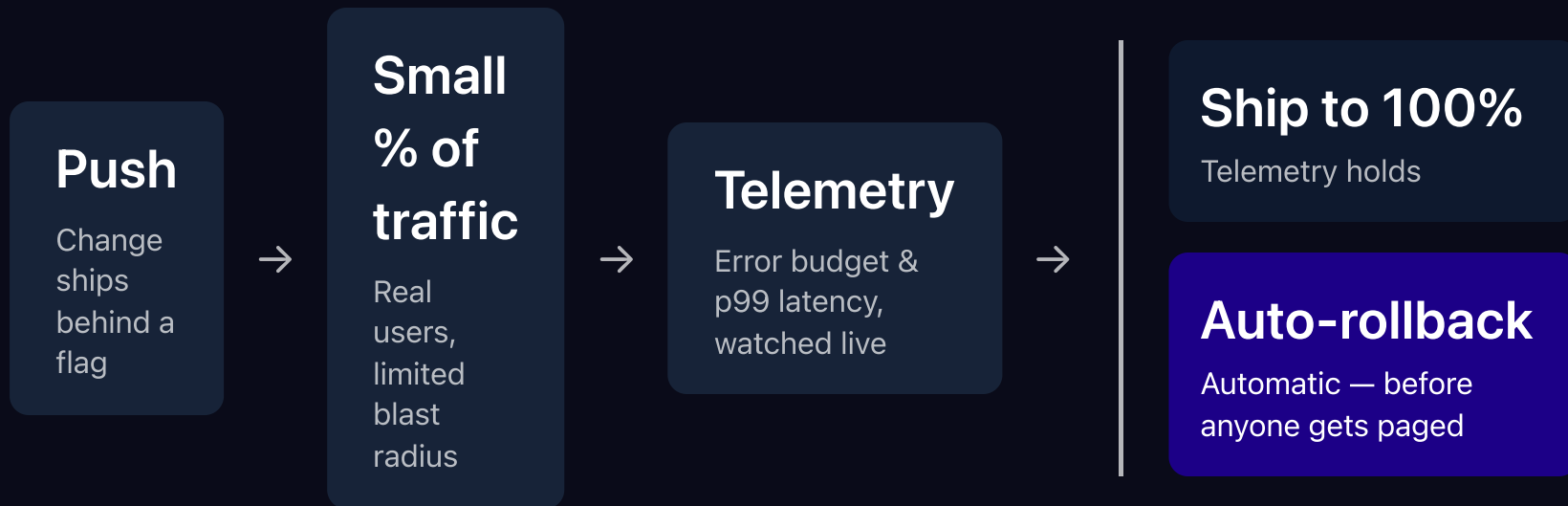
Coverage 80–95%

Authoring takes hours

Payback in ~3 months



Progressive delivery



The cost of skipping it

+80%

more outages, without a
verification layer

+46%

more quality impact, without a
verification layer

Non-users vs. users of a deterministic verification
tool - Sonar 2026

The positive contrast

Coinbase restructured its engineering interviews around directing and verifying AI output - not writing code from scratch.

The AI Capabilities Model - DORA 2025

01

Clear AI stance

A stated, communicated position on AI use.

02

Healthy data ecosystems

Data that's well-governed at the source.

03

AI-accessible data

Internal data AI tooling can actually reach.

05

Small batches

Working in small, reviewable increments.

06

User-centric focus

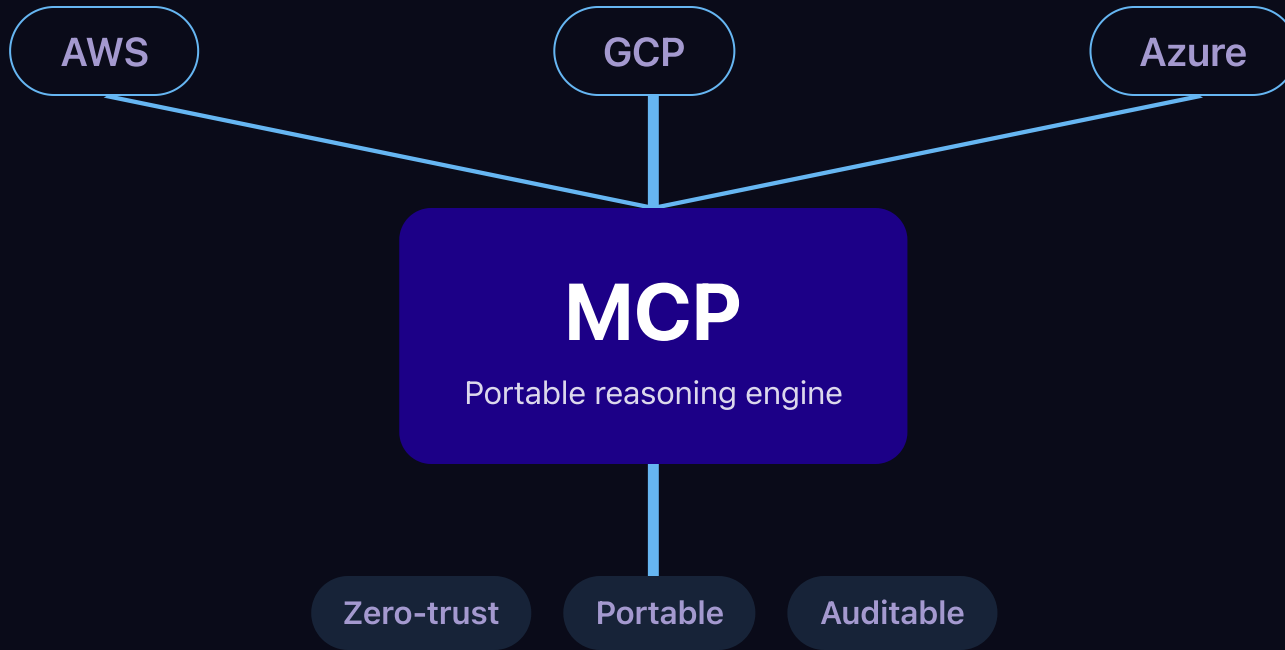
Genuinely centered on the user, not the roadmap.

07

Quality internal platforms

Platforms worth building on.

One protocol, any cloud



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Successful AI adoption is a systems problem, not a tools problem.

DORA, 2026

The Roadmap

Three steps to take home

01

Pick a high-velocity, non-critical surface

Not billing, not auth. Internal tools, a dev dashboard. Put the J-curve dip somewhere it can't hurt you.

02

Instrument the right things

Main-branch throughput, success rate vs. a 90% benchmark, recovery time vs. 60 minutes - not lines of code. As we just saw, vanity metrics can hide the paradox at trillion-token scale.

03

Budget for the tuition, up front

Tell your CFO before the dip arrives that you expect one. Spend political capital holding the line, not explaining yourself after the fact.



Reinvest on purpose, not by accident

By accident

The "free headcount" DORA describes
gets absorbed into more feature
pressure

You stay the implementer

On purpose

Reinvested in quality, security, and
paying down technical debt

You become the systems architect and
quality governor

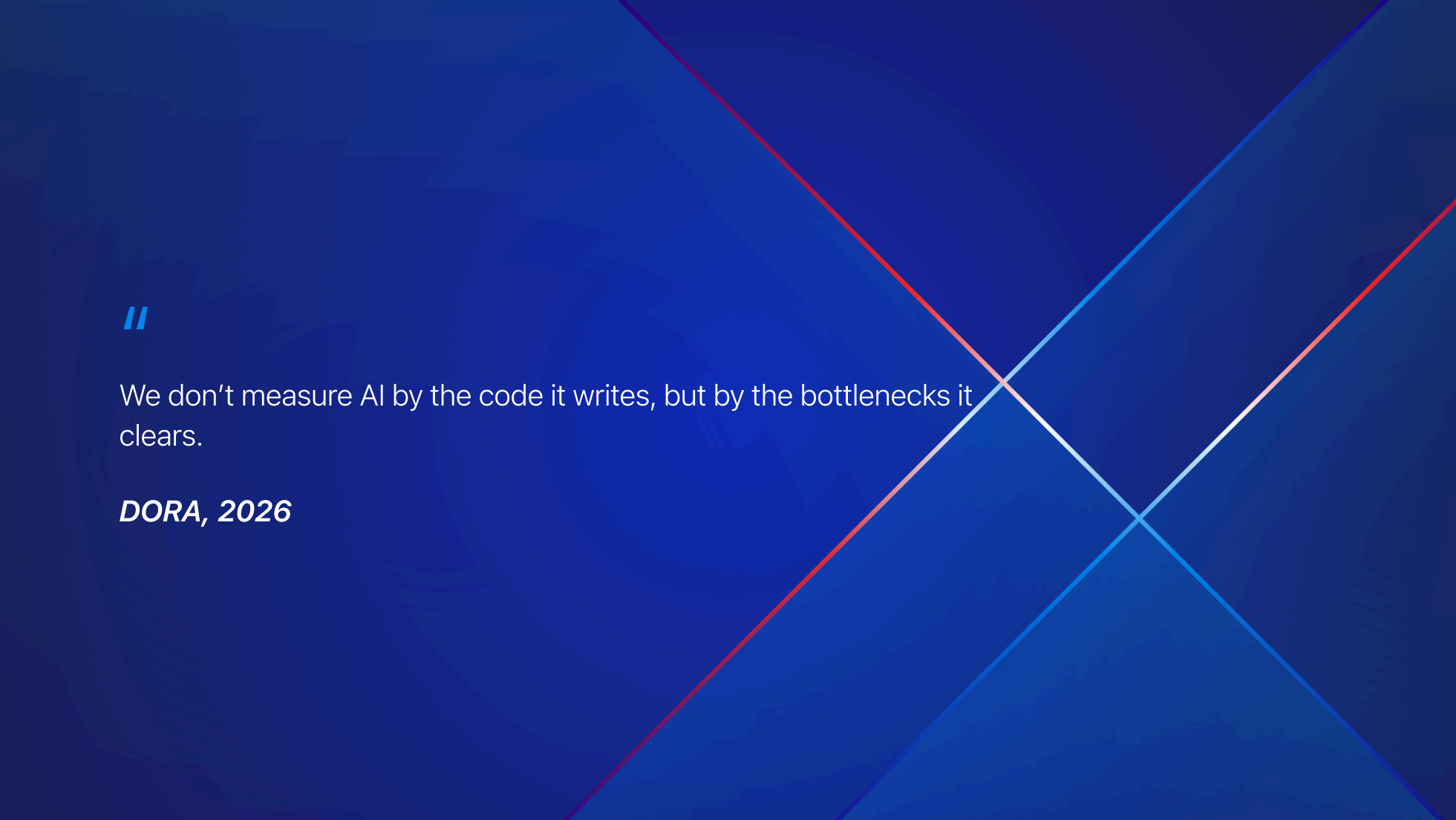


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Code is not an asset. Code is a liability.

Software Engineering at Google

via DORA 2026



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We don't measure AI by the code it writes, but by the bottlenecks it clears.

DORA, 2026

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Thank You

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Slides: <https://github.com/jerdog/talk-dont-panic-agentic-devops>